Your universe at the best real-matter fit: point core plus a baryon-only Sedov–Taylor shell

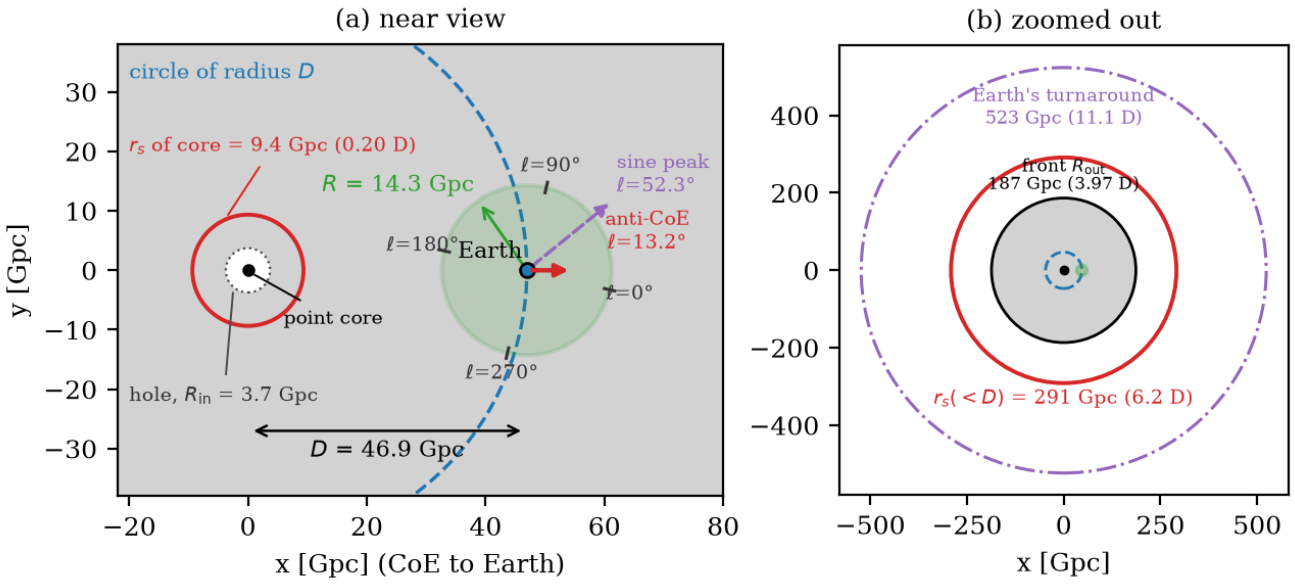


Figure 1: To scale, in Gpc with equal axes. Grey is the shell of constant density $\Omega_s = 0.05$ (baryons only), running from $R_{\rm in} = 3.7$ Gpc to the front at $R_{\rm out} = 187$ Gpc; white inside $R_{\rm in}$ is the hole, and the black dot is the point core (scale 0.23 Gpc). (a) The dashed blue circle of radius D runs through Earth, the green disc is the vision shell of radius R, and sky directions are Galactic longitude with $\theta = \ell - \ell_0$. The red circle is the Schwarzschild radius $r_s = 2GM/c^2$ of the core alone, 9.4 Gpc ($0.20\,D$). (b) Zoomed out to show the front, r_s of all the mass inside D (291 Gpc, $6.2\,D$), and Earth's turnaround radius (523 Gpc). The mass inside D has $r_s > D$. The model is Newtonian, and Earth moves outward at $10.3\,c$, so it has not collapsed. The fit is bound, with $(v/v_{\rm esc})^2 = 0.76$, and Earth turns around at $11.1\,D$ after 34 ages. The mass inside that radius has $r_s \approx 17{,}700$ Gpc, so turnaround falls inside r_s, and in general relativity the region would recollapse. The same holds for any uniform sphere with $\Omega = 0.05$ larger than about 19 Gpc, including the matter of a standard cosmology.

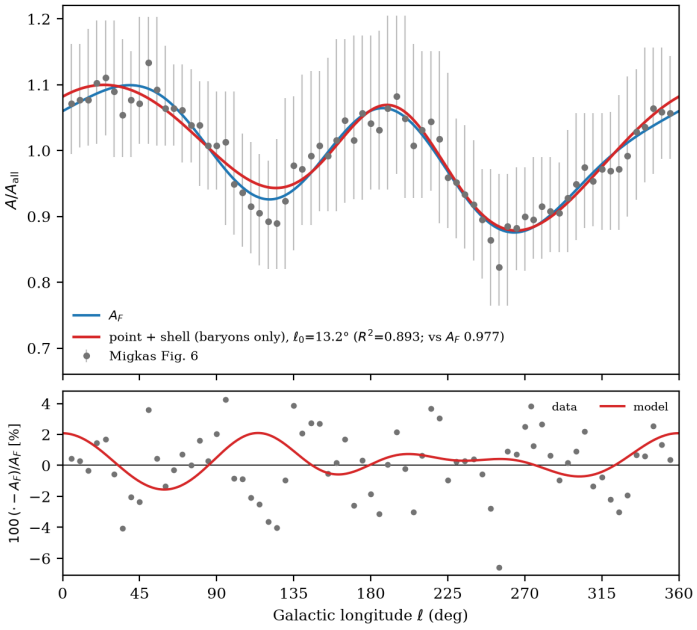


Figure 2: Migkas points with error bars, A_F, and your model A for the point core plus baryon-only shell, normalised to the sky mean. The lower panel shows data and model as percent differences from A_F.

Table 1: Best-fit values in your sheet's cells, integrated model (G real, from $0.01\,t_0$, fixed in space, $H_0 = 70$), with density contrast across the visible sphere $\leq 2\%$ and ℓ_0 free. B27 is all the gravitating mass inside D. The CoE mass is unlabelled; only the shell is set to the baryon density.

Cell	Quantity	Best fit
C23	D/R	3.283
D23	L_c/D (core size)	0.0050 (lower bound)
B29	R	4.4127×10^{23} km (14.30 Gpc)
B28	D	1.4486×10^{24} km (46.95 Gpc)
B30	L_c	7.243×10^{21} km (0.235 Gpc)
B27	$M(<D)$, total	6.052×10^{54} kg $= 3.04 \times 10^{24}\ M_\odot$
	M_c, point core	1.943×10^{53} kg $= 9.77 \times 10^{22}\ M_\odot$
	$M_c/M_{\rm crit}(D)$	0.00166
	$M_{\rm shell}(<D)$	5.857×10^{54} kg
BA57	Amp	−0.0588
BA59	φ	0.889 rad
	ℓ_0	13.21°
	Ω_s (baryons only)	0.05
	ρ_s	4.60×10^{-28} kg m^{-3}
	$R_{\rm in}$	3.73 Gpc ($0.079\,D$)
	$R_{\rm out}$	186.6 Gpc ($3.97\,D$)
	R_b (matter edge)	$= R_{\rm out}$
	r_s, point core	9.35 Gpc ($0.199\,D$)
	r_s, all mass $< D$	291 Gpc ($6.20\,D$)
	$(v/v_{\rm esc})^2$	0.757 (bound)
	Age t_0	13.8 Gyr
	δ across sphere	0 (shell uniform there)
	R^2 vs 71 Migkas points	0.893
	R^2 vs A_F	0.977

The shell theorem, the cored alternative and your thin shell

By Newton's shell theorem, matter arranged symmetrically about the CoE pulls a source at radius r as if everything inside r sat at the centre, and matter outside r exerts no net pull. Three rules follow for your picture. First, any symmetric distribution inside $D-R$ acts on every source we see as a single point mass at the CoE, so its profile cannot be measured, only its total mass (and, in the integrated model, the depth of the potential the sources climbed out of). Second, symmetric matter beyond $D+R$ pulls on nothing we see, so its density is free. Third, only matter between $D-R$ and $D+R$ changes across the visible sphere, so for the density there to be uniform, a constant-density shell must be at least $2R$ thick, from $D-R$ to $D+R$. Inside such a shell $g = \frac{4\pi}{3}G\rho_s r + G(M_c - M_{\rm hole})/r^2$ with $M_{\rm hole} = \frac{4\pi}{3}\rho_s R_{\rm in}^3$. The linear term is isotropic, so in the field today the net central mass $M_c - M_{\rm hole}$ must be positive, or the tidal pattern reverses.

The time-integrated model softens that last rule. The sources start at 1% of their present scale and move out through the shell, which is fixed in space. The force always uses the mass inside each particle's radius at that moment, so a source spends most of the age inside the empty hole, where only M_c pulls, and enters the shell only in roughly the last R/D of the age (15–30%). Because the core acted alone for most of that history, your thin shell with a core fits with the usual $\ell_0 \approx 13°$ even though its net central mass is negative (Table 2). With no core the pattern turns by about 90° ($\ell_0 = 105$–$108°$), as the negative-mass argument predicts, and the fit weakens.

The cored profile $\rho \propto 1/(1+(r/L)^2)$, the equal-mass-shell picture with $\rho \propto 1/r^2$ far out, is a live alternative. On its own its contrast is about $4R/D$, so it meets $\delta \le 2\%$ only at $D = 200R$, where $R^2 = 0.809$. With the LCDM-equivalent background the limit is met at $D = 10.8R$ ($D = 155$ Gpc, $L = 1.6$ Gpc), with $R^2 = 0.843$. Allowing D/R above the earlier cap of 10 raised this from 0.830. The cored fits are 13.0–13.6 Gyr old, and bound only with the $\Omega = 0.3$ background. Your thin shell hugging the visible sphere gives the best R^2 of any model so far, 0.909 (0.987 against A_F), at $D = 94$ Gpc. It is bound and 13.8 Gyr old, but it needs $\Omega_s = 1.01$, about 20 times the baryon density. At baryon density it fits as well as the best real-matter model ($R^2 = 0.893$) but leaves Earth unbound. Moving the shell edges out by $0.1R$ changes R^2 by less than 0.001.

Table 2: Alternatives to the best real-matter fit, all with ℓ_0 free, sky-normalised and your $p = 1$ convention. Lengths in Gpc. M is all gravitating mass inside D. Ω is the shell or background density; "LCDM" is the LCDM-equivalent 0.3 and "baryons" is 0.05. Thin shells run from $D-R$ to $D+R$, so their contrast across the sphere is zero. r_s/D uses $r_s = 2GM(<D)/c^2$.

Model	Matter	D/R	D	Shape [Gpc]	M [$M_\odot$]	Ω	ℓ_0	$R^2_{\rm Migkas}$	$R^2_{A_F}$	δ [%]	$(v/v_{\rm esc})^2$	Age	r_s/D
point + ST shell (best real matter)	baryons	3.28	47	L_c 0.23; shell 3.7–187	3.04×10^{24}	0.05	13	0.893	0.977	0	0.76	13.8	6.2
cored $1/(1+(r/L)^2)$ + background	LCDM	10.84	155	L 1.56	1.06×10^{26}	0.30	15	0.843	0.919	2	0.73	13.0	65
cored + background	baryons	48.44	693	L 6.9	8.92×10^{27}	0.05	15	0.815	0.889	2	3.92	13.5	1,233
cored alone	none	199.98	2860	L 28.7	6.21×10^{29}	0.00	15	0.809	0.882	2	12.82	13.6	20,795
thin shell + core	free	6.55	94	shell 79–108	1.86×10^{26}	1.01	13	0.909	0.987	0	0.90	13.8	190
thin shell + core	baryons	3.47	50	shell 35–64	2.34×10^{24}	0.05	13	0.893	0.977	0	10.56	13.9	4.5
thin shell only	free	8.28	118	shell 104–133	3.19×10^{26}	1.08	108	0.830	0.905	0	1.30	13.8	258
thin shell only	baryons	3.77	54	shell 40–68	2.71×10^{24}	0.05	105	0.332	0.368	0	13.30	13.9	4.8

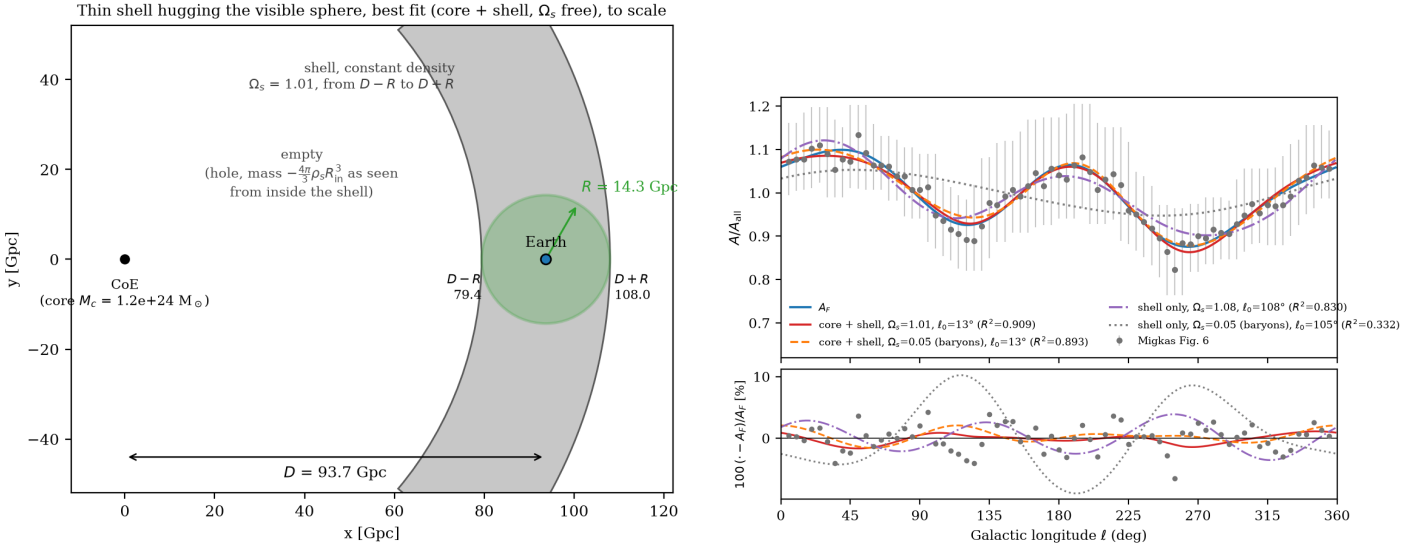


Figure 3: Left: your thin shell, to scale, at its best fit (core plus shell with Ω_s free). The shell of constant density runs from $D-R$ to $D+R$, and the interior is empty apart from the core. Right: thin-shell fits against the Migkas points and A_F. With no core, the pattern turns by about 90°.